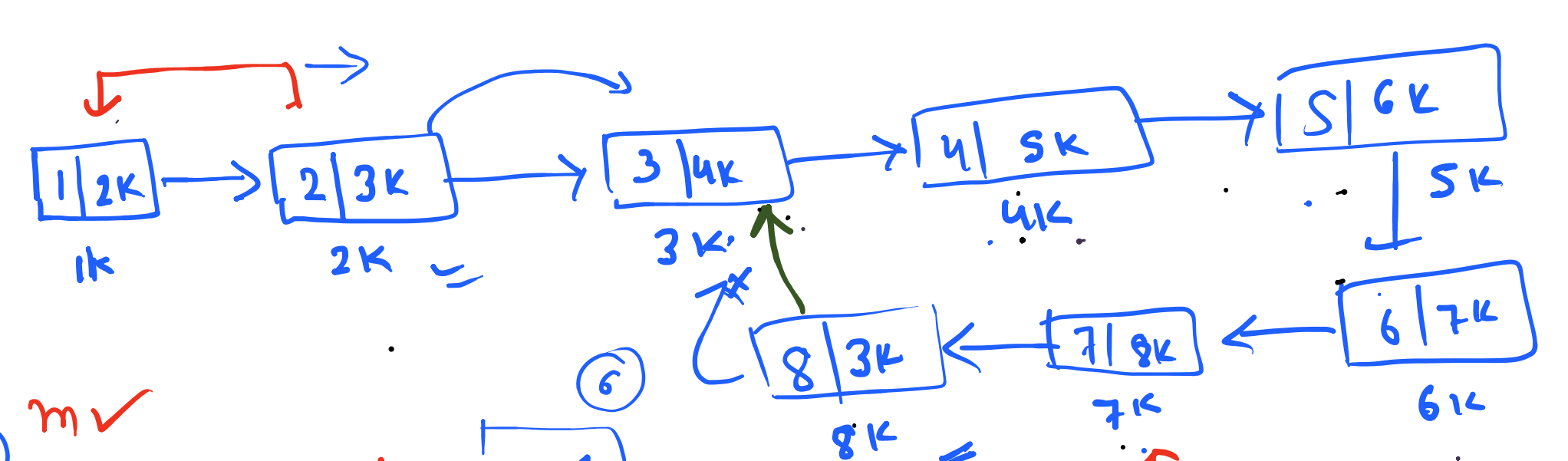
Lec-20

13/07/26 10:27 AM

tail ✓

```
public void CreateCycle() {
    tail.next = getNode(2);
}
```


1 | 2K
1K
2 | 3K
2K
3 | 4K
3K
4 | 5K
4K
5 | 6K
5K
6 | 7K
6K
7 | 8K
7K
8 | 3K
8K
6

m ✓

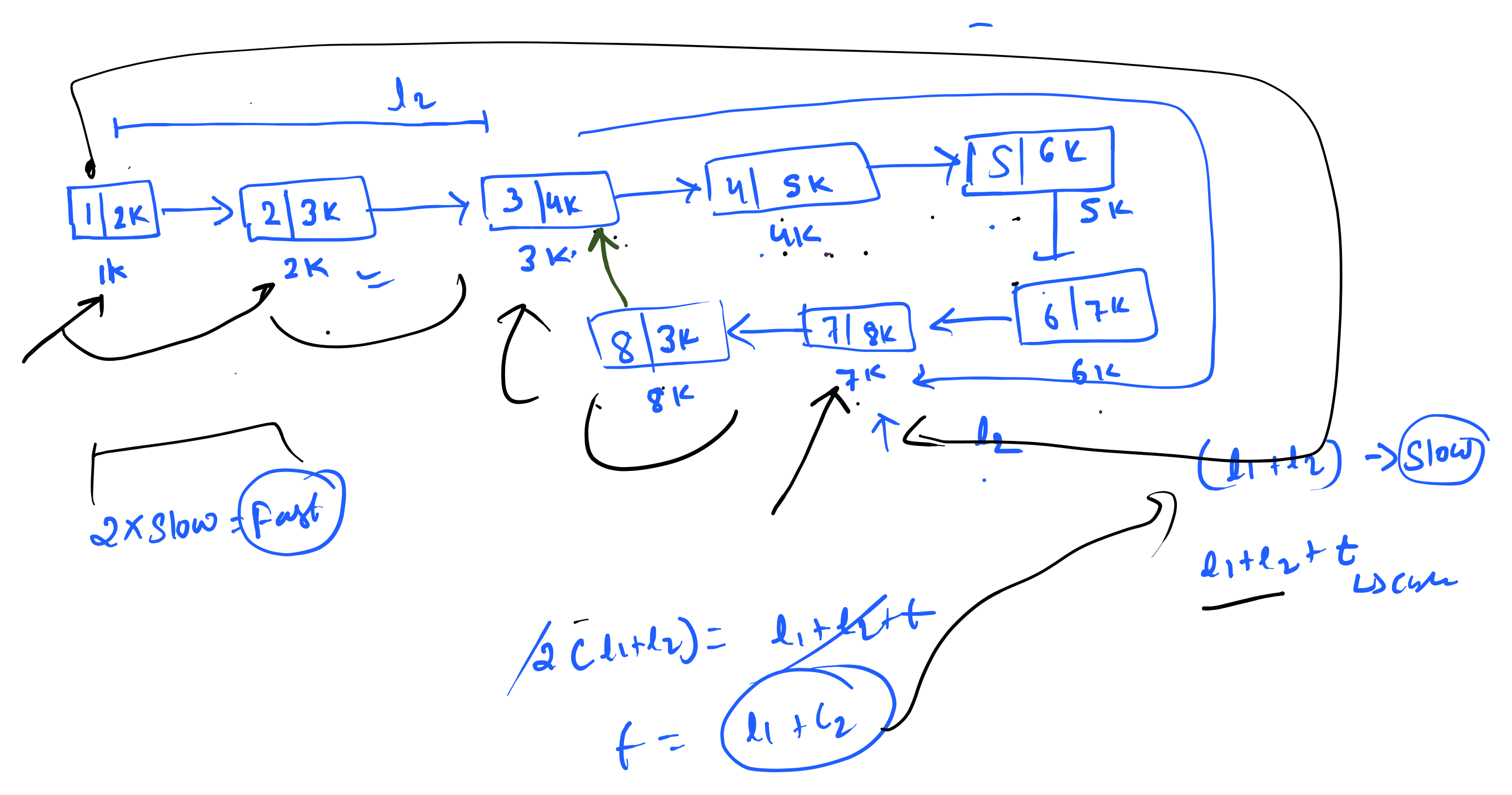
```
(n) (i) Node temp = meet , c = 1
    while(temp.next != meet){
        c++
        temp = temp.next
    }

(n) (ii) Node fast = head
    for(i=0 i<c i++){
        fast = fast.next
    }

(iii) (n) while(slow.next != fast.next){
        slow = slow.next
        fast = fast.next
    }
    fast.next = null
```

```
Node start = head
while(start != null){
    Node temp = meet
    while(temp.next != meet){
        if(temp.next == start){
            temp.next = null
        }
        temp = temp.next
    }
    start = start.next
}
```

N ✓
N+M ✓

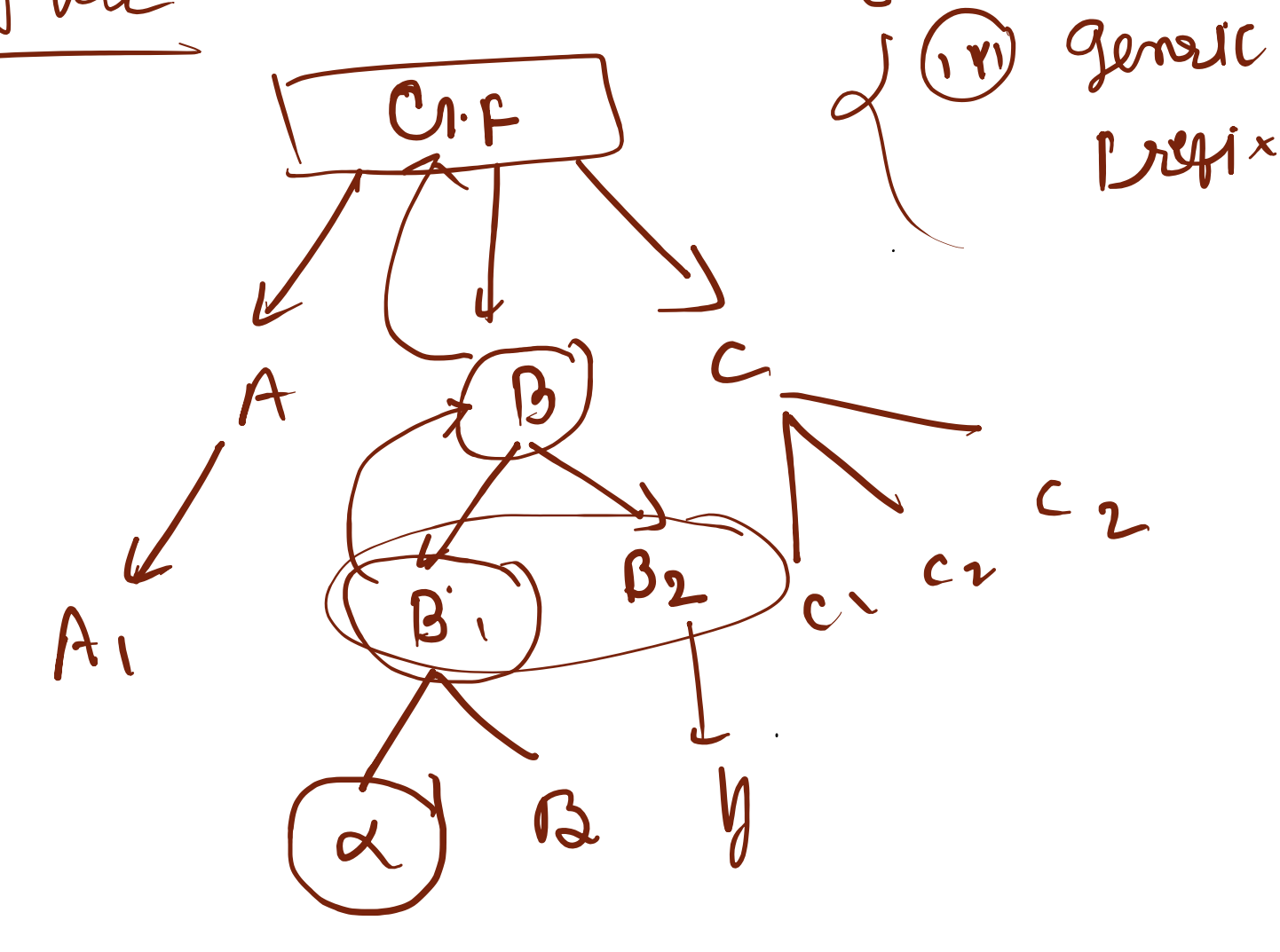

l2
1 | 2K
1K
2 | 3K
2K
3 | 4K
3K
4 | 5K
4K
5 | 6K
5K
6 | 7K
6K
7 | 8K
7K
8 | 3K
8K
l2
2×slow = fast
(l1+l2) -> slow
2(l1+l2) = l1+l2+t
f = l1+l2
l1+l2+t LsCycle

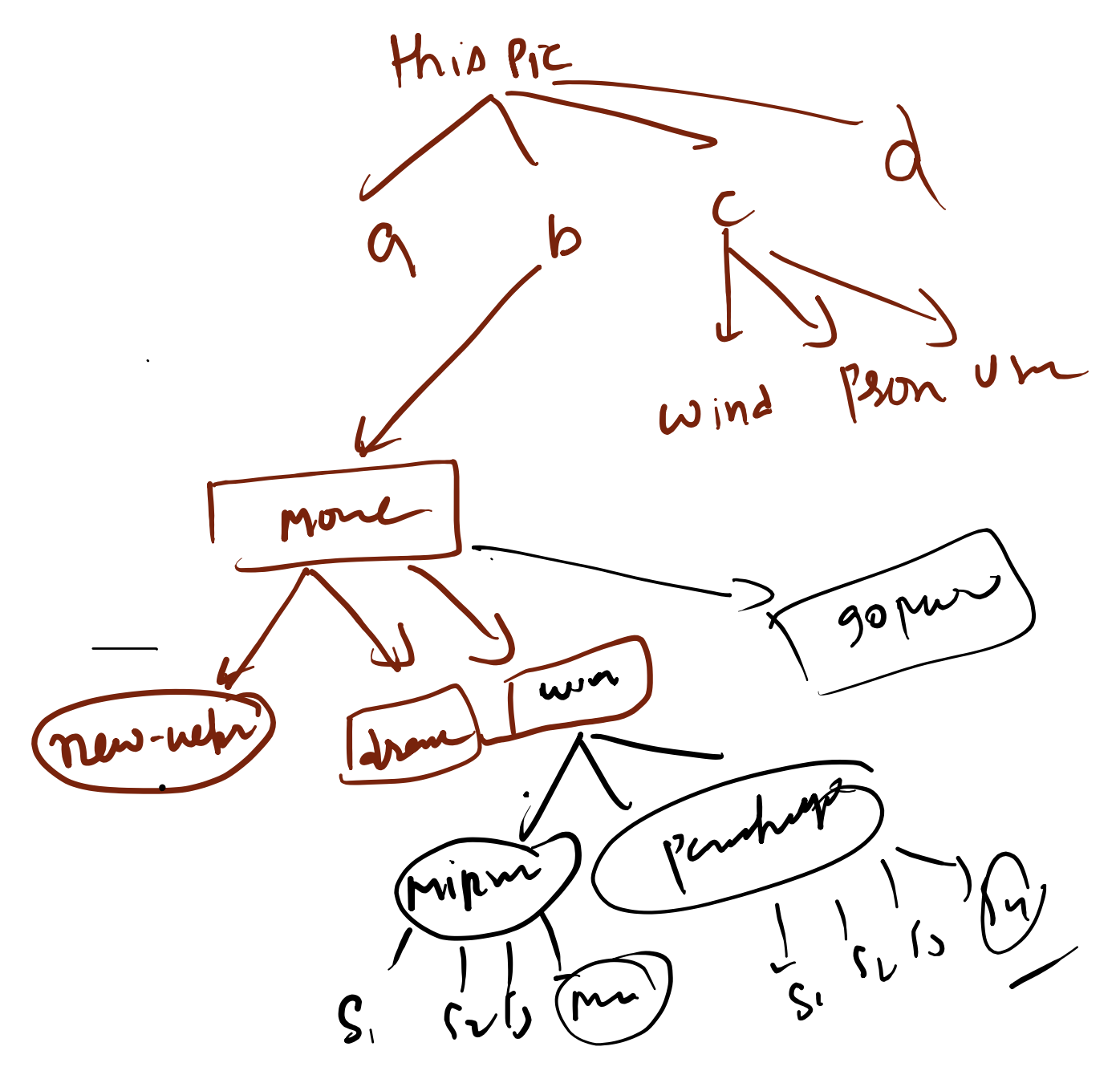
Tree

Thi

① Family tree

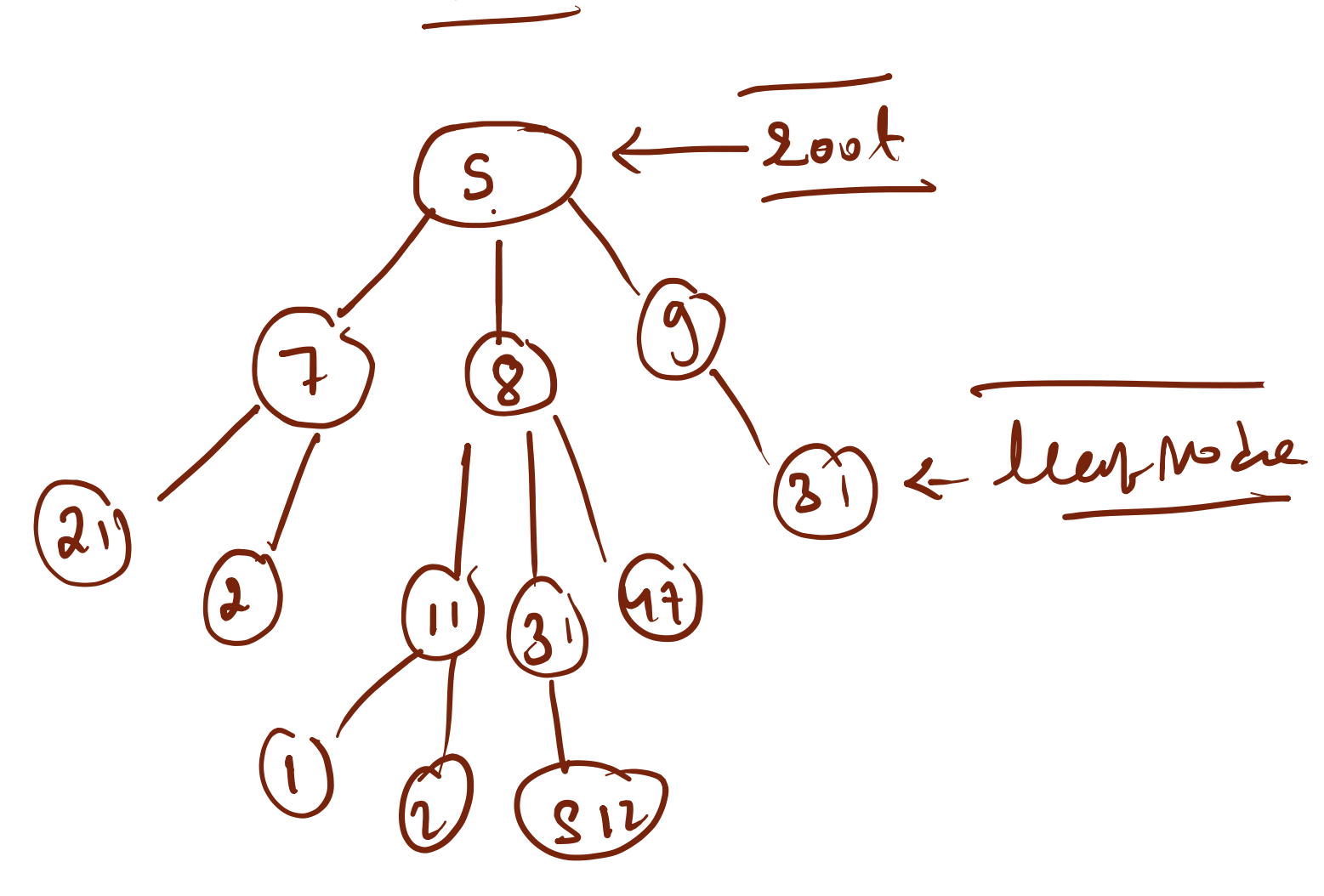
(i) Binary Tree
(ii) Binary Search Tree
(iii) Generic Tree
Prefix Tree


G.F
A
B
C
A1
B1
B2
C1
C2
C2
α
β
y


This PC
a
b
c
d
wind
Person vm
Movie
new-web
drama
war
90 pnw
mipur
punchingw
S1
S2 S3
mu
S1
S2 S3
S4

G.T

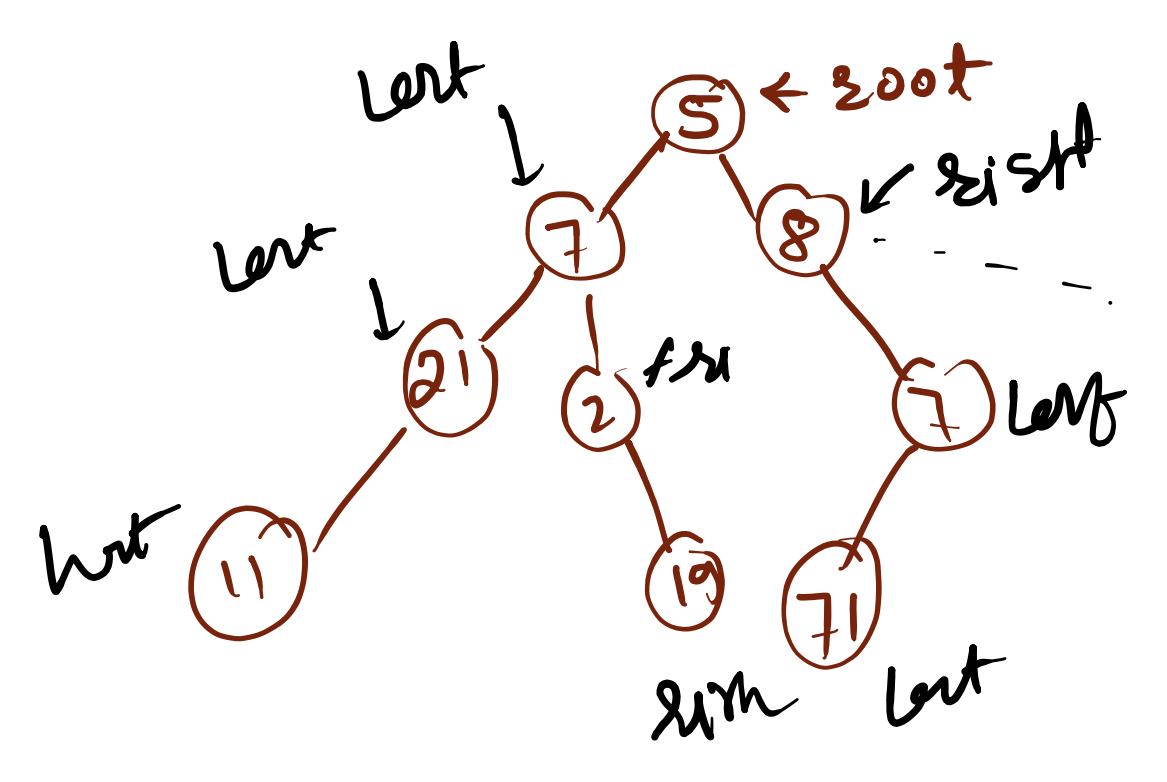
parent-child


S
root
7
8
9
21
2
11
31
47
31
leaf Node
1
2
812

B.T & B.S.T

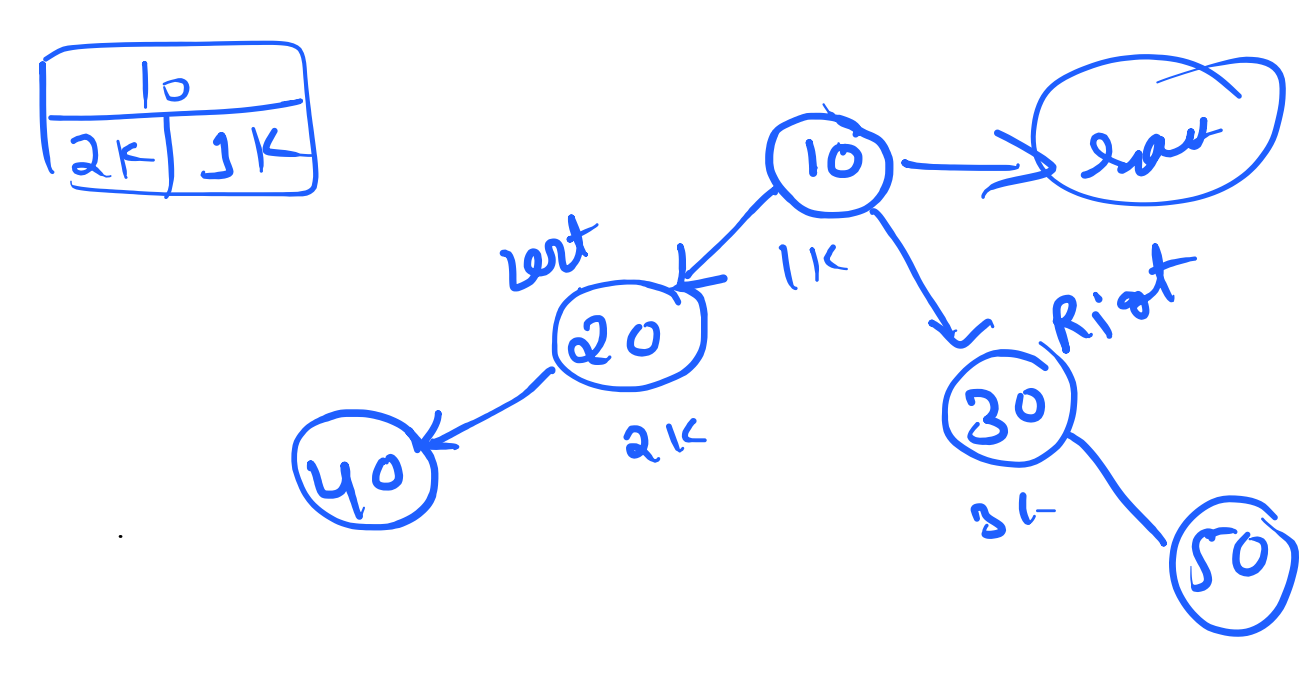
① हम दो हमारे दो

② दो बच्चा मिट्ठी सिर्फ
उससे ज्यादा बकवास


Left
Left
Left
5
root
7
8
right
21
2
7
left
11
19
71
right
Left

number boolean

[10 T 20 T 40 F F F T 30 F T 50 F F]

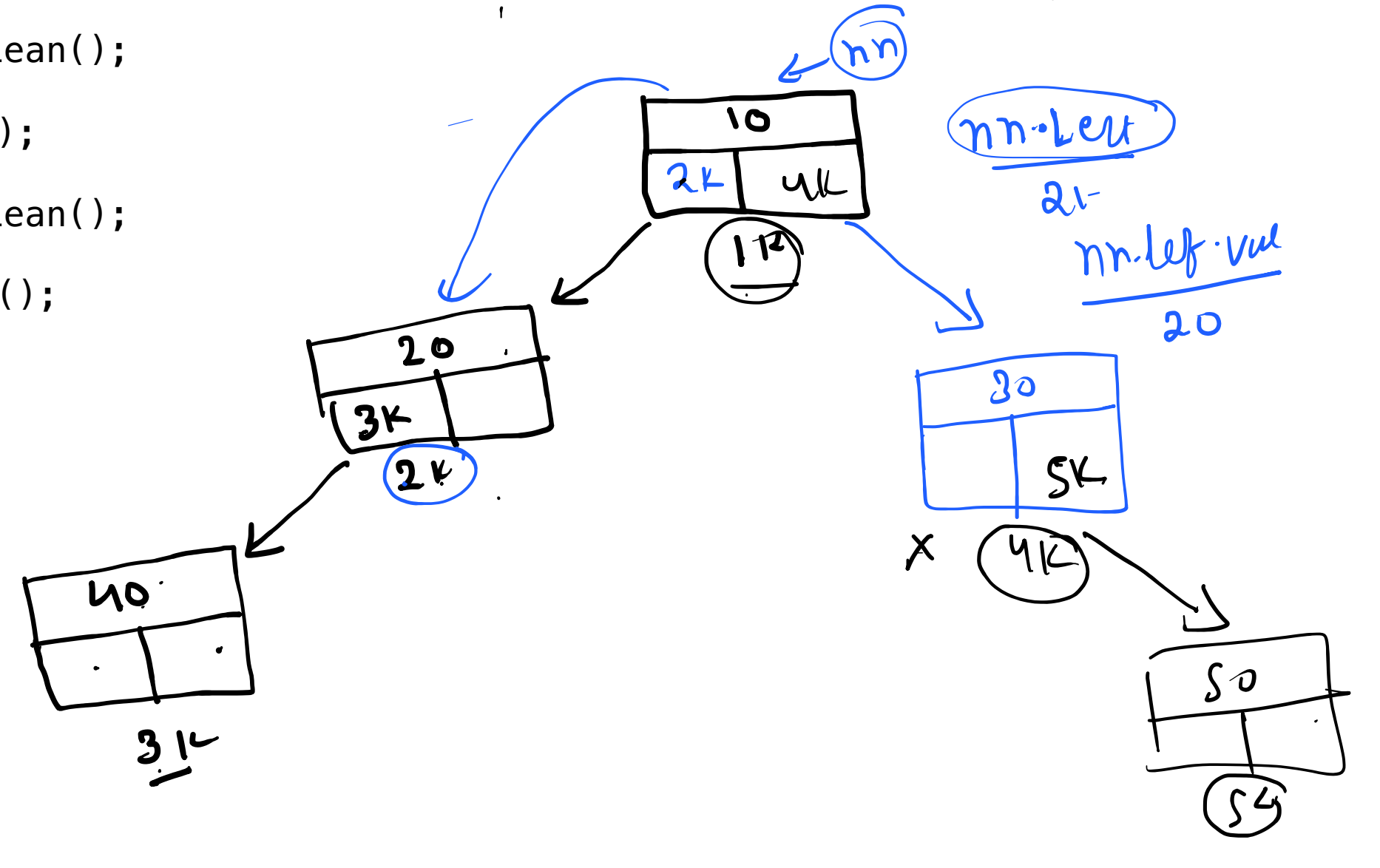

10
2K | 1K
10
1K
root
Left
20
2K
40
Right
30
3K
50

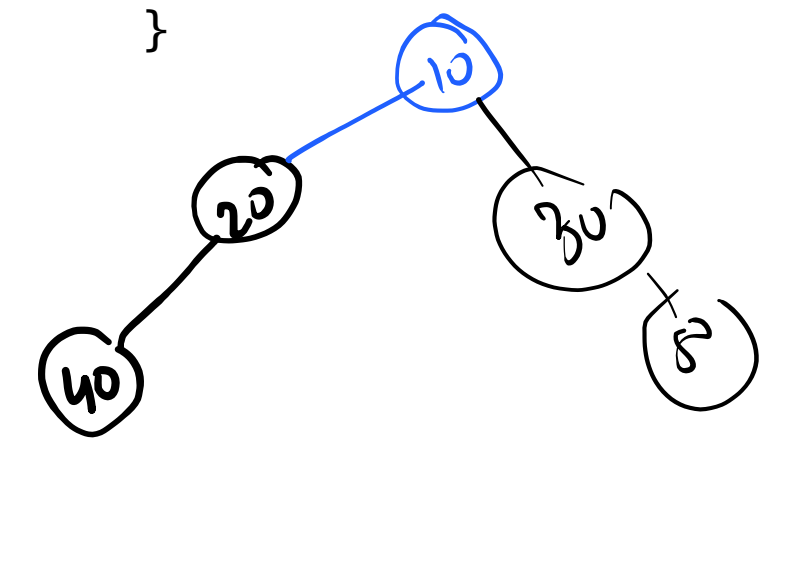
root
Left subtree
Right subtree

Class Node {
int val
Node left
Node Right

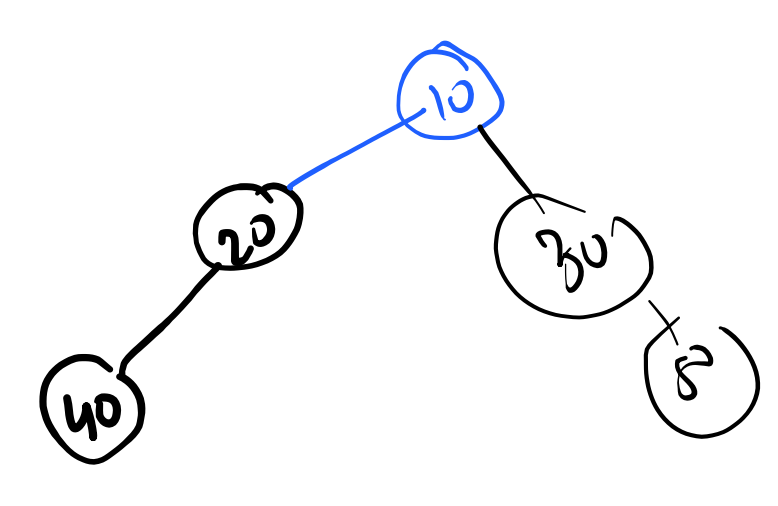
```
private Node createTree() {
    int item = sc.nextInt();
    Node nn = new Node(item);
    boolean hlc = sc.nextBoolean();
    if (hlc) {
        nn.left = createTree();
    }
    boolean hrc = sc.nextBoolean();
    if (hrc) {
        nn.right = createTree();
    }
    return nn;
}
```

[10 T 20 T 40 F F F T 30 F T 50 F F]


nn
10
2K | 4K
1K
nn.left
2K
nn.left.val
20
20
3K
2K
40
3K
30
5K
4K
50
5K

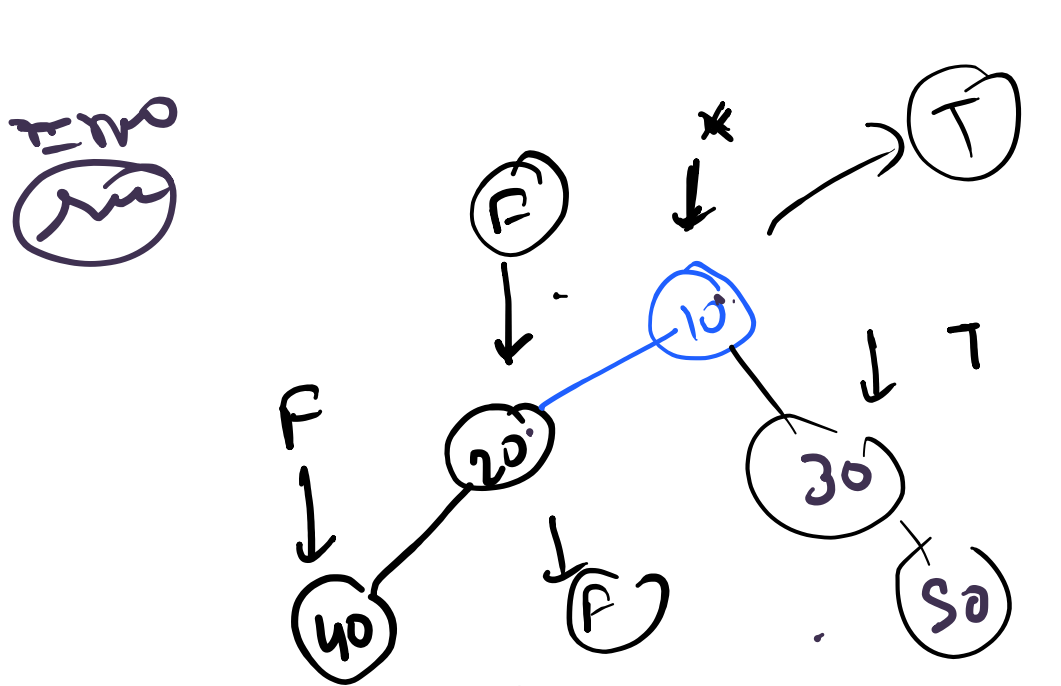

10
20
30
40
50

←10→

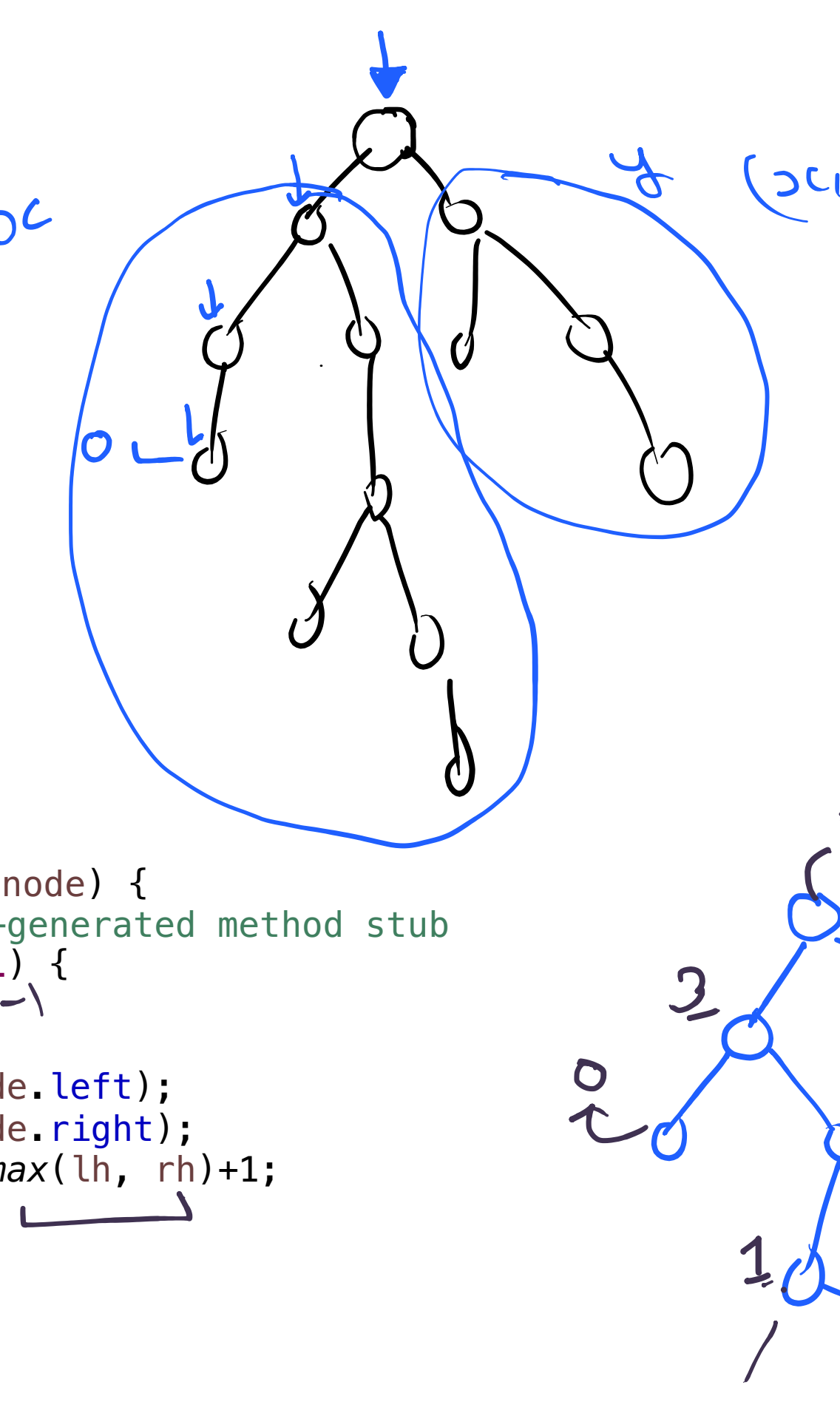

10
20
30
40
50

20 ← 10 → 30
40 ← 20 → .
. ← 40 → .
. ← 30 → 50
. ← 50 → .

```
public void Display(Node nn) {
    String s = "<--"+nn.val+"-->";
    if(nn.left!=null) {
        s=nn.left.val+s;
    }
    else {
        s="."+s;
    }
    if(nn.right!=null) {
        s=s+nn.right.val;
    }
    else {
        s=s+".";
    }
    System.out.println(s);
    Display(nn.left);
    Display(nn.right);
}
```


nn
Error
T
F
10
T
F
20
30
40
F
50
F
F

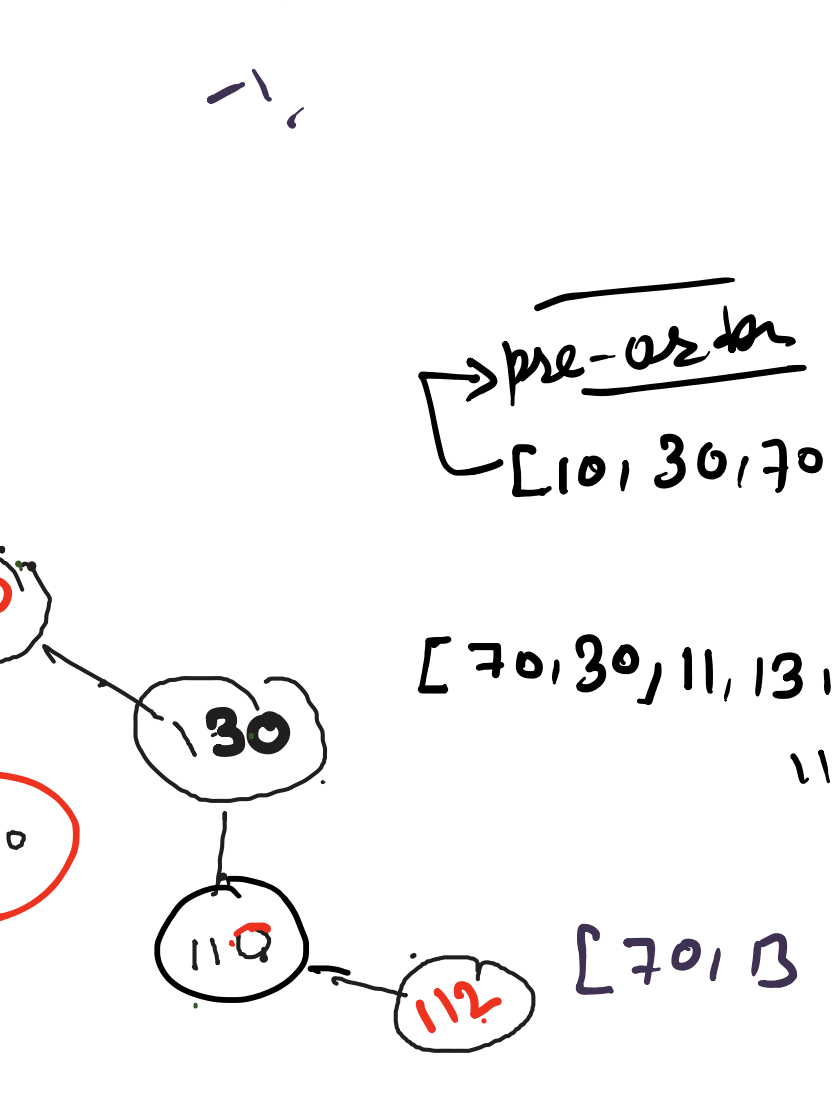
20 ← 10 → 30
40 ← 20 → .
. ← 40 → .
. ← 30 → 50
. ← 50 → .

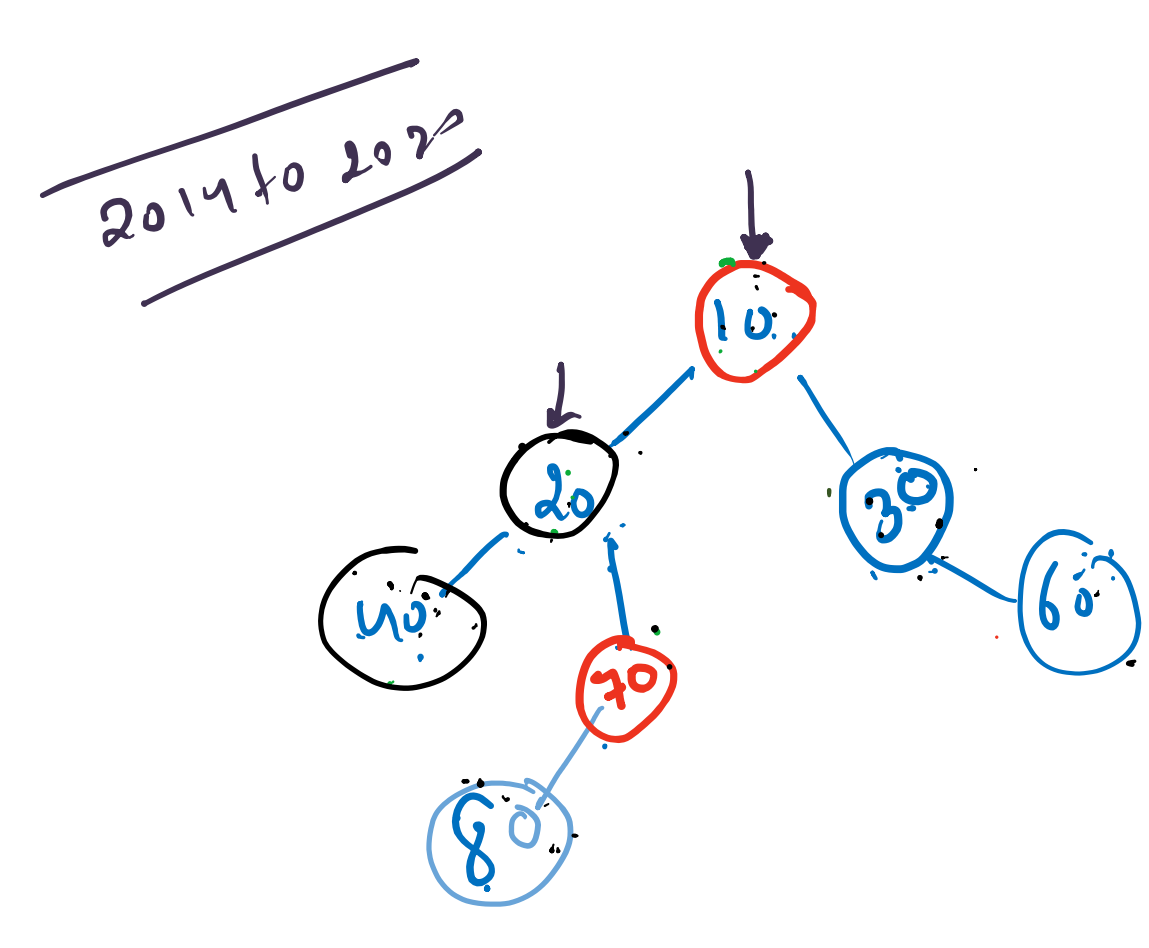


maximum distance b/w root to leaf

0 → 1
→ 0

```
private int ht(Node node) {
    // TODO Auto-generated method stub
    if(node==null) {
        return 0;-1
    }
    int lh=ht(node.left);
    int rh=ht(node.right);
    return Math.max(lh, rh)+1;
}
```



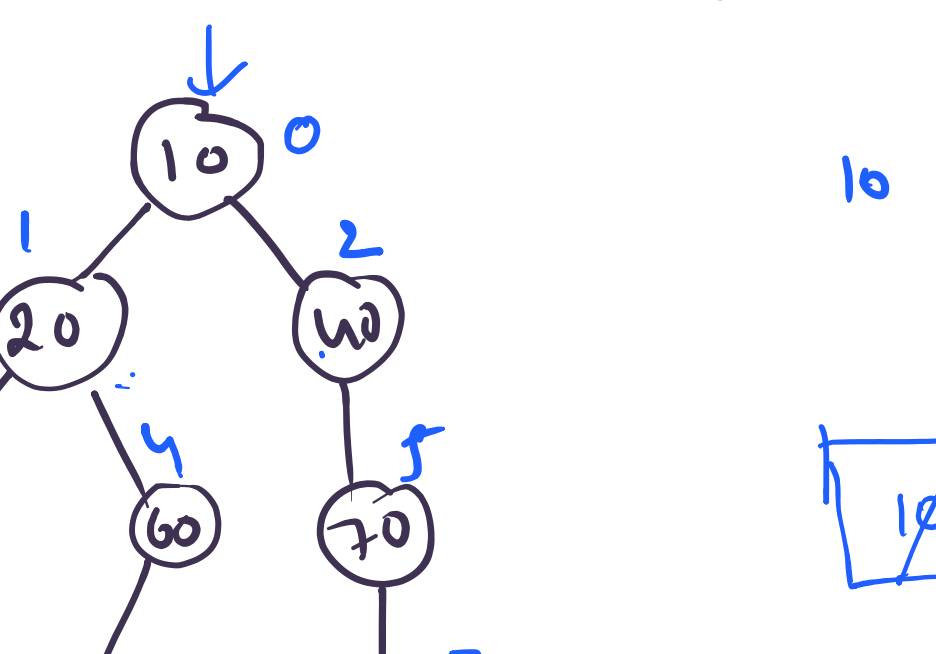

10
140
80
30
170
120
70
50
18
130
11
12
110
112
13

pre-order
[10, 30, 70, 50, 11, 12, 13, 18, 140, 170, 80, 120]
13-, 110, 112)

[70, 30, 11, 13, 12, 50, 18, 10, 170, 140, 120, 180,
110, 112, 130)

[70, 13, 12, 11, 18, 50, 30, 170, 120, 112, 110]
130, 80, 140, 10)

2014 to 2020

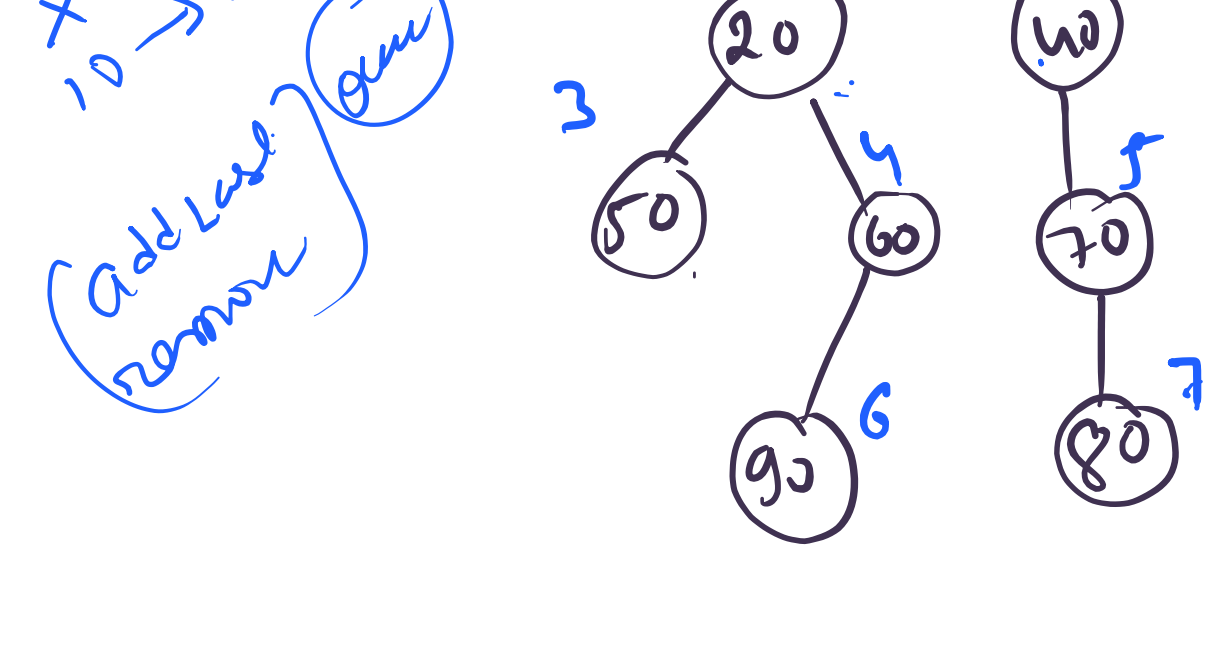

10
20
30
40
70
60
80
addF
removeL
addL

① Pre-order → Root, left, Right
② in-order → left, Root, Right
③ Post-order → left, Right, Root

[10, 20, 40, 70, 80, 30, 60]

40 20 80 70 10 30, 60

[40, 80, 70, 20, 60, 30, 10]


get
x
10 → 20 → 30 → 40
addLast
removeF
gun
10
0
20
1
40
2
3
4
5
50
60
70
6
7
90
80

10 20 40 50 60 90 70 80

10	20	40	50	60	70	90	80

10 20 - 40 50 60 70 90 80